



**NATIONAL INSTITUTE OF BUSINESS MANAGEMENT
SCHOOL OF COMPUTING AND ENGINEERING**

DSE 26.1F

**DATABASE MANAGEMENT 01 - COURSEWORK
SOLAR CONTRACTOR
COMPANY MANAGEMENT SYSTEM**

GROUP

CODSE261F-019

CODSE261F-031

CODSE261F-061

CODSE261F-044

Acknowledgement

We extend our sincere gratitude to the National Institute of Business Management (NIBM) for providing the resources necessary for this project. Special thanks to our lecturer Mr. Shashika for his guidance in Data management enabling us to translate theoretical concepts into a real-world hardware application.

We would also like to express our heartfelt appreciation to [Sennon Energy Solution (Pvt) Ltd for their generous support, cooperation, and for allowing us the opportunity to study and implement this system within their organization. Their domain insights and operational support were instrumental in the successful completion of this coursework.

Table Of Content

1. List of tables
2. List of figures
3. Chapter 1 - introduction
4. Chapter 2 - analysis
5. Chapter 3 - design
 1. Conceptual level
 2. Logical level
 3. Physical level
6. Chapter 4 - conclusion

List Of Tables

Table 1 - Entity selection and attributes

Table 2 - Relation between entities

List Of Figures

Figure 1 - Site visite

Figure 2 - EER diagram

Figure 3 - Schema Diagram

Chapter 1 - introduction

This is implementation of a Database system for a Solar Constructor Company system following standard database design steps and guidelines and making the current system more efficient, secure and reliable.

As steps follows:

1. Analyse
 - Determine what should go in the database and how the information should use
 - Mainly forces on the identifying the business requirements, business rules, adding necessary security performance
2. Conceptual design
 - Goal it to create a user and business focused model of the database
 - Identifying the necessary entities and relation between them using Enhanced Entity-Relationship Diagram (EER).
3. Logical design
 - Convert the conceptual model into a well designed schema based specific paradigm
 - Identifying tables, columns, data types, keys, constraints, and normalizing data to eliminate data to eliminate redundancy
4. Physical design
 - Create the logical schema to a specific Database management System for optimal performance and Storage
 - Choosing specific DBMS data types, creating indexing strategies, partitioning, file storage organization, and hardware resource allocation
5. Testing and deploy
 - Find out necessary functionality, performance, and integrity before getting in the live users
 - Change schema to the production environment, importing initial data, and opening the database for live system integration.

Chapter 2 - Analysis



Figure 1

- **Role Specialization** :Employees are mainly divided into three categories - Engineer, Technician ,or Driver. An employee must not hold a couple of roles at the same time.
- **Solar System Tracking**: Each solar System is uniquely identified by System code and must maintain key technical specifications, including Wattage, Power, Dimensions and base Price.
- **Supplier Sourcing**: Needed systems and components are supplied by registered suppliers, tracked by the company registration number and company name
- **Driver Authorization** : Only employees known as the Driver and hold recorded driver's license are authorized to drive company vehicles.
- **Customer Identification**: Corporate or individual clients differentiate using Customer ID, also using details like Phone number, Address and Email for further more clarification.

Entities and Attribute selection

Entity	Definition	Attributes
Employee	Shows the staff of the company that they hired to achieve a specific task.	Emp_ID (PK), Name, a Status, Phone_num, DOB, Address, Email, Tax_ID, Salary
Engineer	Specialized employee subtype that completes tasks related to the field of engineering.	Emp_ID (PK, FK), Type
Technician	Specialized employee subtype that completes tasks related with technical services and repairs.	Emp_ID (PK, FK), Skill_level
Driver	Specialized employee subtype that completes tasks related to operating vehicles.	Emp_ID (PK, FK), Dri_Lic, V_ID (FK)
Dependent	Shows us the family members or people who depend on the employee.	Emp_ID (PK, FK), Name (PK), DOB, Gender, Relation
Customer	Represents the individuals or corporate entities that purchase a service or a system.	Cus_ID (PK), Name, Status, Phone_num, DOB, Address, Email
Order	Symbolize customers placing a purchase request order.	Order_ID (PK, FK), Date, Cus_ID (FK), Sys_ID (FK)
Solar System	Portrays the installed solar systems.	Sys_code (PK), Name, Dimension, Price,

		Warranty, type, Battery_cap, Power, Sup_ID (FK)
Supplier	Shows us the vendor that supplies the solar components.	Sup_ID (PK), Status, Com_name, Com_Reg_Num
Installment	Follows up with the payment installments and services that are linked to the installment.	Ins_ID (PK), Date, Price, Emp_ID (FK), Warranty, Order_ID (FK)
Activity	Portrays the process of the installation and the performance logs.	Sys_code (PK, FK), Date (PK), Time (PK), Flag, Wattage
Vehicle	Represent vehicles that are owned by the company and that are assigned for the drivers.	V_ID (PK), V_type, V_Status

Table 1

Entities and relationships

Total 7 relations

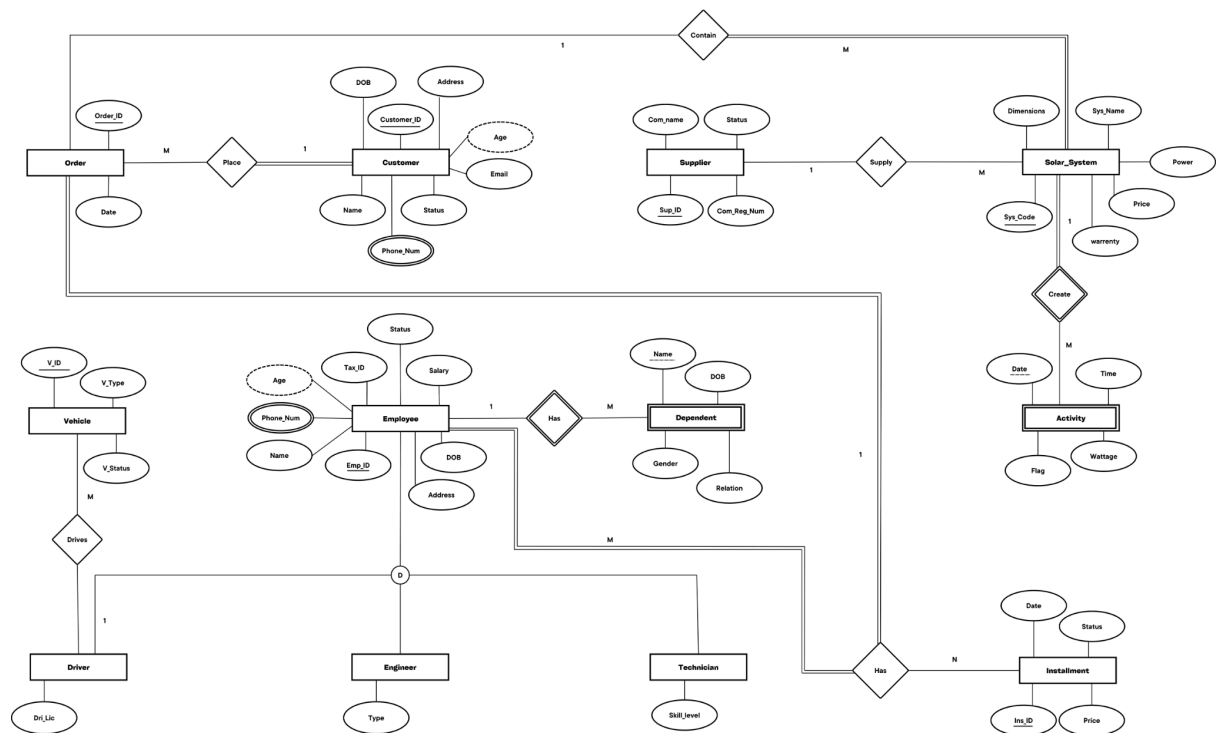
Entities	Relationships	Cardinality
Customer → Order	A single customer can place multiple orders over time, but each individual order belongs to only one specific customer.	1 : M
Order → Solar_System	An order can contain several different products, and a single product can be included in many different customer orders.	M : N
Solar_System → Activity	Whenever an action happens in the system it automatically creates a single log entry in the audit trail. A system can create 1 or many activity logs, each activity generated by only one system.	1 : M
Dependent → Employee	An employee can have 0 or many dependents. A dependent must belong to exactly 1 employee.	1 : M
Supplier → Solar_System	Supplier supplies one or many solar systems. Each system is supplied by one customer.	1 : M
Employee → Installment	Many technicians can be part of multiple installments, multiple employers can work on many installments over time.	M : N
Driver → Vehicle	A driver drives 1 or many vehicles, each vehicle assigned to 1 driver.	1 : M

Table 2

Chapter 3 - Design

1. Conceptual Layer (EER Diagram)

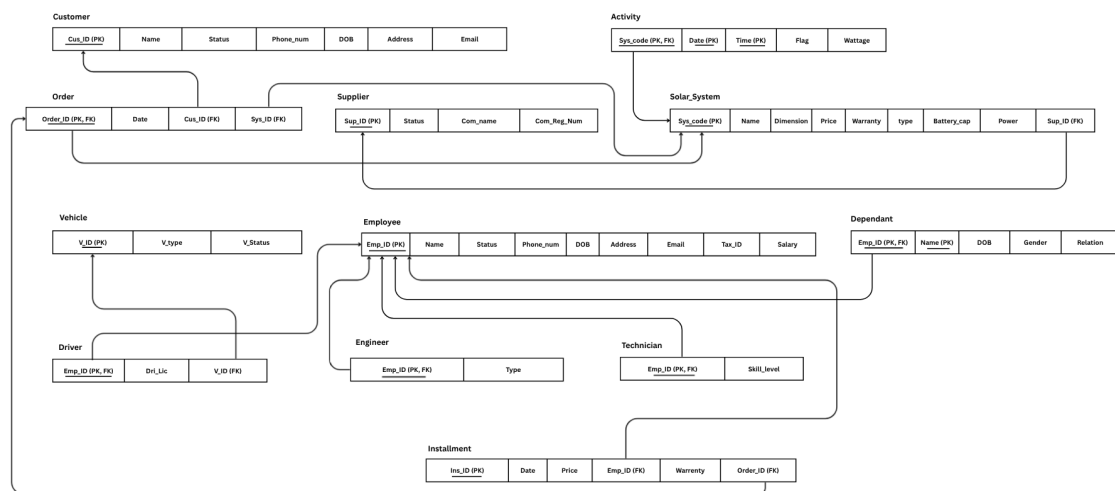
We focus here on visually representing the entities, attributes and relationships using an Enhanced Entity-Relationship (EER) Diagram.



The high resolution graph can be found in: <https://github.com/NeDeSilva/SMS-DB/>

2. Logical Layer (Relational Schema Diagram & Mapping)

We are translating our conceptual EER diagram into a relational schema. Convert our entities into tables that have columns, primary keys (PK), foreign keys (FK). Map the subclasses in our diagram (create separate tables for entities Employee, Engineer, Technician and Driver linked by using the employee_id as foreign key). The Design is already in 3rd Normal Form.



The high resolution graph can be found in: <https://github.com/NeDeSilva/SMS-DB/>

Database Normalization Statement

The database schema satisfies First Normal Form (1NF), Second Normal Form (2NF), and Third Normal Form (3NF) based on the following structural characteristics.

- **First Normal Form (1NF):** All tables contain only atomic (indivisible) values within each column, with no repeating groups or array structures; multi-valued attributes (such as Phone_Num) and derived attributes (such as Age) are transformed into separate relations or calculated dynamically to ensure table atomicity.

- Second Normal Form (2NF): The database meets 1NF requirements, and every non-key attribute is fully functionally dependent on the entire primary key of its respective table, ensuring no partial dependencies exist (particularly in junction tables or entities with composite keys).
- Third Normal Form (3NF): The database meets 2NF requirements, and no non-key attribute is transitively dependent on the primary key, meaning every non-key column depends solely on the primary key and nothing else.

3. Physical Layer (SQL DDL & Constraints)

We Implement our database schema specifically for your targeted RDBMS (MS SQL Server)

All the codes can be found in: <https://github.com/NeDeSilva/SMS-DB/>

Create Database

```
-- check if database exist before creation
-- UTF-8 encoding for full Unicord support

IF NOT EXISTS (
    SELECT 1
    FROM sys.Databases
    WHERE name = N'SENNON_ENERGY'
)
BEGIN
    CREATE DATABASE SENNON_ENERGY
    COLLATE Latin1_General_100_BIN2_UTF8;
END;
GO
```

Insert Data to tables

```
-- 1. Customer
CREATE TABLE Customer (
    Cus_ID VARCHAR(50) PRIMARY KEY,
    Name VARCHAR(100),
    Status VARCHAR(50),
    Phone_num VARCHAR(20),
    DOB DATE,
    Address VARCHAR(255),
    Email VARCHAR(100)
);

-- 2. Supplier
CREATE TABLE Supplier (
    Sup_ID VARCHAR(50) PRIMARY KEY,
    Status VARCHAR(50),
    Com_name VARCHAR(100),
    Com_Reg_Num VARCHAR(100)
);

-- 3. Vehicle
CREATE TABLE Vehicle (
    V_ID VARCHAR(50) PRIMARY KEY,
    V_type VARCHAR(50),
    V_Status VARCHAR(50)
);

-- 4. Employee
CREATE TABLE Employee (
    Emp_ID VARCHAR(50) PRIMARY KEY,
    Name VARCHAR(100),
    Status VARCHAR(50),
    Phone_num VARCHAR(20),
    DOB DATE,
    Address VARCHAR(255),
    Email VARCHAR(100),
    Tax_ID VARCHAR(50),
    Salary DECIMAL(10, 2)
);

-- 5. Solar System
```

```

CREATE TABLE Solar_System (
    Sys_code VARCHAR(50) PRIMARY KEY,
    Name VARCHAR(100),
    Dimension VARCHAR(50),
    Price DECIMAL(10, 2),
    Warranty VARCHAR(50),
    Type VARCHAR(50),
    Battery_cap VARCHAR(50),
    Power VARCHAR(50),
    Sup_ID VARCHAR(50),
    FOREIGN KEY (Sup_ID) REFERENCES Supplier(Sup_ID)
);

-- 6. Activity
CREATE TABLE Activity (
    Sys_code VARCHAR(50),
    Date DATE,
    Time TIME,
    Flag VARCHAR(50),
    Wattage VARCHAR(50),
    PRIMARY KEY (Sys_code, Date, Time),
    FOREIGN KEY (Sys_code) REFERENCES Solar_System(Sys_code)
);

-- 7. Order
CREATE TABLE `Order` (
    Order_ID VARCHAR(50) PRIMARY KEY,
    Date DATE,
    Cus_ID VARCHAR(50),
    Sys_ID VARCHAR(50),
    FOREIGN KEY (Cus_ID) REFERENCES Customer(Cus_ID),
    FOREIGN KEY (Sys_ID) REFERENCES Solar_System(Sys_code)
);

-- 8. Driver (Subtype of Employee)
CREATE TABLE Driver (
    Emp_ID VARCHAR(50) PRIMARY KEY,
    Dri_Lic VARCHAR(50),
    V_ID VARCHAR(50),
    FOREIGN KEY (Emp_ID) REFERENCES Employee(Emp_ID),
    FOREIGN KEY (V_ID) REFERENCES Vehicle(V_ID)
);

```

```

-- 9. Engineer (Subtype of Employee)
CREATE TABLE Engineer (
    Emp_ID VARCHAR(50) PRIMARY KEY,
    Type VARCHAR(50),
    FOREIGN KEY (Emp_ID) REFERENCES Employee(Emp_ID)
);

-- 10. Technician (Subtype of Employee)
CREATE TABLE Technician (
    Emp_ID VARCHAR(50) PRIMARY KEY,
    Skill_level VARCHAR(50),
    FOREIGN KEY (Emp_ID) REFERENCES Employee(Emp_ID)
);

-- 11. Dependant
CREATE TABLE Dependant (
    Emp_ID VARCHAR(50),
    Name VARCHAR(100),
    DOB DATE,
    Gender VARCHAR(20),
    Relation VARCHAR(50),
    PRIMARY KEY (Emp_ID, Name),
    FOREIGN KEY (Emp_ID) REFERENCES Employee(Emp_ID)
);

-- 12. Installment
CREATE TABLE Installment (
    Ins_ID VARCHAR(50) PRIMARY KEY,
    Date DATE,
    Price DECIMAL(10, 2),
    Emp_ID VARCHAR(50),
    Warranty VARCHAR(50),
    Order_ID VARCHAR(50),
    FOREIGN KEY (Emp_ID) REFERENCES Employee(Emp_ID),
    FOREIGN KEY (Order_ID) REFERENCES `Order` (Order_ID)
);

```

Queries

```
USE SENNON_ENERGY;
GO

-- SECTION 1: BASIC SELECT QUERIES (5 Queries)

-- Query 1: Retrieve all active customers and their primary
contact details
SELECT Cus_ID, Name, Phone_num, Email, Address
FROM Customer
WHERE Status = 'Active';

-- Query 2: Display all solar systems with a price greater than
$5,000, sorted by price (highest first)
SELECT Sys_code, Name, Type, Price, Power
FROM Solar_System
WHERE Price > 5000.00
ORDER BY Price DESC;

-- Query 3: List all active employees, including their
title/salary and tax ID
SELECT Emp_ID, Name, Status, Tax_ID, Salary
FROM Employee
WHERE Status = 'Active';

-- Query 4: Show high-wattage activity readings flagged as 'Peak'
or 'Warning'
SELECT Sys_code, Date, Time, Flag, Wattage
FROM Activity
WHERE Flag IN ('Peak', 'Warning');

-- Query 5: Retrieve all available vehicles for driver assignment
SELECT V_ID, V_type, V_Status
FROM Vehicle
WHERE V_Status = 'Available';

-- SECTION 2: ADVANCED SELECT QUERIES (5 Queries)

-- Query 6: Find total revenue per customer order count for
customers who spent more than $10,000 total
SELECT Order_ID, COUNT(Ins_ID) AS Total_Installments, SUM(Price)
```

```

AS Total_Spent
FROM Installment
GROUP BY Order_ID
HAVING SUM(Price) > 10000.00
ORDER BY Total_Spent DESC;

-- Query 7: Calculate average salary by employee role type for
-- subtype tables with more than 1 employee
SELECT e.Type AS Engineer_Type, COUNT(e.Emp_ID) AS Engineer_Count,
AVG(emp.Salary) AS Avg_Salary
FROM Engineer e
JOIN Employee emp ON e.Emp_ID = emp.Emp_ID
GROUP BY e.Type
HAVING COUNT(e.Emp_ID) >= 2
ORDER BY Avg_Salary DESC;

-- Query 8: Identify suppliers providing more than 1 solar system
-- model with an average model price over $5,000
SELECT Sup_ID, COUNT(Sys_code) AS Models_Offered, AVG(Price) AS
Avg_System_Price
FROM Solar_System
GROUP BY Sup_ID
HAVING COUNT(Sys_code) > 1 AND AVG(Price) > 5000.00
ORDER BY Avg_System_Price DESC;

-- Query 9: Count total dependents per employee for employees
-- having more than 1 dependent
SELECT Emp_ID, COUNT(Name) AS Total_Dependents
FROM Dependant
GROUP BY Emp_ID
HAVING COUNT(Name) > 1
ORDER BY Total_Dependents DESC;

-- Query 10: Find total recorded activity logs per solar system
-- with more than 1 logged event
SELECT Sys_code, COUNT(*) AS Total_Logs
FROM Activity
GROUP BY Sys_code
HAVING COUNT(*) > 1
ORDER BY Total_Logs DESC, Sys_code ASC;

```

```
-- SECTION 3: JOIN QUERIES (3 Queries)
```

```
-- Query 11: Join Customer, Order, and Solar_System to view
complete order details
```

```
SELECT
    o.Order_ID,
    o.Date AS Order_Date,
    c.Name AS Customer_Name,
    c.Email AS Customer_Email,
    s.Name AS Solar_System_Name,
    s.Type AS System_Type,
    s.Price
FROM `Order` o
JOIN Customer c ON o.Cus_ID = c.Cus_ID
JOIN Solar_System s ON o.Sys_ID = s.Sys_code
ORDER BY o.Date DESC;
```

```
-- Query 12: Join Driver, Employee, and Vehicle to map drivers to
their assigned vehicles
```

```
SELECT
    e.Emp_ID,
    e.Name AS Driver_Name,
    e.Phone_num,
    d.Dri_Lic,
    v.V_ID,
    v.V_type,
    v.V_Status
FROM Driver d
JOIN Employee e ON d.Emp_ID = e.Emp_ID
JOIN Vehicle v ON d.V_ID = v.V_ID
ORDER BY e.Name ASC;
```

```
-- Query 13: Join Installment, Order, Employee, and Solar_System
to track installment service assignments
```

```
SELECT
    i.Ins_ID,
    i.Date AS Installment_Date,
    i.Price AS Amount_Paid,
    i.Order_ID,
    e.Name AS Technician_In_Charge,
    s.Name AS Installed_System
FROM Installment i
```

```
JOIN `Order` o ON i.Order_ID = o.Order_ID  
JOIN Employee e ON i.Emp_ID = e.Emp_ID  
JOIN Solar_System s ON o.Sys_ID = s.Sys_code  
ORDER BY i.Date ASC;
```

Insert Data

Only Customer table is shown

```
INSERT INTO Customer (Cus_ID, Name, Status, Phone_num, DOB,
Address, Email) VALUES
('CUS001', 'John Doe', 'Active', '555-0101', '1985-03-15', '123
Elm St, Springfield', 'john.doe@email.com'),
('CUS002', 'Jane Smith', 'Active', '555-0102', '1990-07-22', '456
Oak St, Springfield', 'jane.smith@email.com'),
('CUS003', 'Alice Johnson', 'Inactive', '555-0103', '1978-11-05',
'789 Pine St, Metropolis', 'alice.j@email.com'),
('CUS004', 'Bob Brown', 'Active', '555-0104', '1982-01-30', '101
Maple Ave, Gotham', 'bob.brown@email.com'),
('CUS005', 'Charlie Davis', 'Active', '555-0105', '1995-09-12',
'202 Birch Rd, Star City', 'charlie.d@email.com'),
('CUS006', 'Diana Prince', 'Active', '555-0106', '1988-04-18',
'303 Cedar Ln, Gateway', 'diana.p@email.com'),
('CUS007', 'Evan Wright', 'Inactive', '555-0107', '1992-12-01',
'404 Walnut St, Coast City', 'ewright@email.com'),
('CUS008', 'Fiona Gallagher', 'Active', '555-0108', '1996-06-25',
'505 Spruce St, Bludhaven', 'fiona.g@email.com'),
('CUS009', 'George Clark', 'Active', '555-0109', '1975-08-14',
'606 Ash Dr, Keystone', 'gclark@email.com'),
('CUS010', 'Hannah Abbott', 'Active', '555-0110', '1993-02-28',
'707 Beech Ct, Hub City', 'hannah.a@email.com');
```

Permission Levels

```
USE SENNON_ENERGY;
GO

-- 1. Create Database Roles
CREATE ROLE AdminRole;
CREATE ROLE ManagerRole;
CREATE ROLE EmployeeRole;
GO

-- 2. Grant Permissions on the default schema (dbo)
-- Admin permissions
GRANT CONTROL TO AdminRole;

-- Manager permissions
GRANT SELECT, INSERT, UPDATE, DELETE ON SCHEMA::dbo TO
ManagerRole;

-- Employee permissions
GRANT SELECT ON SCHEMA::dbo TO EmployeeRole;
GO

-- 3. Create Server Logins (Authentication Level)
CREATE LOGIN admin_user WITH PASSWORD = 'StrongAdminPassword123!';
CREATE LOGIN manager_user WITH PASSWORD =
'StrongManagerPassword123!';
CREATE LOGIN employee_user WITH PASSWORD =
'StrongEmployeePassword123!';
GO

-- 4. Create Database Users (Database Level)
CREATE USER admin_user FOR LOGIN admin_user;
CREATE USER manager_user FOR LOGIN manager_user;
CREATE USER employee_user FOR LOGIN employee_user;
GO

-- 5. Add Users as Members of Database Roles
ALTER ROLE AdminRole ADD MEMBER admin_user;
ALTER ROLE ManagerRole ADD MEMBER manager_user;
ALTER ROLE EmployeeRole ADD MEMBER employee_user;
GO
```

Conclusion

Project Overview and Achievements

For Sennon Energy Solution (Pvt) Ltd we successfully designed and implemented centralized, customized relational Database Management System (DBMS) . The primary objective was to replace fragmented tracking systems with a structured database solution capable of managing employee role specializations, inventory, and company operations efficiently.

Through these systematic engineering process, our system successfully fulfilled these key requirements:

- **Requirements Analysis**

We mapped out the operational rules, especially including the disjoint role enforcement (Engineers, Technicians, Drivers), driver licensing authorization, and dependent associations.

- **Conceptual Modeling**

We developed an Enhance Entity-Relationship (EER) model to show complex relationships, superclass/superclass hierarchies and business constraints.

- **Logical Design & Normalization**

We Translate our EER model into relational schemas and we apply normalization up to 3rd Normal Form (3NF) because it eliminates data redundancy and anomalies.

- **Physical Implementation**

We executed SQL Data Definition Language (DDL) scripts by combining primary keys, foreign key cascading rules, check constraints and performance indexes to ensure data integrity and query performance.

Key System Benefits

There are significant Advantages for Sennon Energy Solution (Pvt) Ltd through our newly designed database system

1.Data Consistency and Integrity

Because of centralizing all entity information it prevents conflicting records while strict schema constraints guarantee valid data inputs across the operational processes.

2.Operational Security

When it comes to operational security there are access rules and role-based specialization constraints to ensure authorization rules (verifying authorized drivers before assigning company vehicles).

3.Scalability and Performance

We optimised our relational structures and targeted database by indexing to support higher query throughput as system data grows.

Future Enhancements

Since our database works perfectly right now ,but we can add these features to our database in future updates

- **Automated Audit Logging**

It works like an automatic logbook. If someone updates inventory, installs new gear, or changes employee info, a permanent record is created that no one can delete or alter.

- **Cloud Database Migration**

By Hosting our database on cloud infrastructure (AWS RDS or Azure SQL) it enables remote accessibility for field technicians and real time mobile reporting.

- **Role-Based Access Control (RBAC)**

By implementing fine grained user privilege tiers directly within the DBMS help to enforce our security boundaries between operational staff and executive administrators.